

CY 524: Basic and Applied Electrochemistry

Course Code	CY 524
Course Name	Basic and Applied Electrochemistry
L-T-P-C	3-0-0-3 Distribution : L-T-P-C
Prerequisites	Undergraduate level Physical Chemistry courses
Intended for	M.Sc Chemistry, MTech, PhD
Approval	50th BoA (Jul 2023)

Course Contents

- **Interfacial Electrochemistry:** Introduction and over view of electrochemical processes, Basic electrochemical thermodynamics, free energy, Nernst Equation, half reaction and electrochemical potentials, formal potentials, liquid junction potential, Faradic and non-Faradaic processes, electrode-electrolyte interface, electrical double layer, polarizable and non-polarizable interfaces, Pourbaix diagram, thermodynamics of batteries. (8 Hours)
- **Kinetics of Electrode reactions and Measurements:** Essentials of electrode reactions, Butler Volmmer Model for electrode kinetics, one step-one electron process through potential energy diagram, standard rate constants, symmetry factor and transfer coefficients, Tafel slops, equilibrium condition and exchange current, mechanistic criteria; diffusion, activation phenomena, electron transfer theories, Marcus Theory, electrochemical transport process. (8 Hours)
- **Techniques for electrochemical methods:** Current-potential relationship; methods of measurement of kinetic parameters; over potential, electrochemical Experiments and variables in electrochemical cells, reference electrode, three electrode cell, supporting electrolyte, steady state and potential step techniques; polarography; cyclic voltammetry; chronomethods; convective diffusion systems: rotating disc and ring disc electrodes; microelectrodes; impedance techniques - concepts and applications, Equivalent Circuit Dynamics, differential pulse voltammetry, square wave voltammetry, linear sweep voltammetry. (14 Hours)
- **Application of Electrochemistry:** Pourbaix Diagram and relation to electrochemistry Fundamentals of batteries: primary, secondary, reserve batteries; solid state and molten solvent- batteries; heterogeneous catalysis, sensor, fuel cells, photo-electrochemical solar cells and conversion of solar energy, Corrosion – fundamentals and applications. (10 Hours)
- **Laboratory/practical/tutorial Modules:** 0

Text Books

1. J.O.M Bokris and A.K.N, Reddy, **Modern Electrochemistry**, Volume 1 and 2, Plenum Press, 1998.
2. A.J. Bard and L.R. Faulkner, **Electrochemical Methods**, 2nd Edition, John Wiley and Son, 2001.

References

1. Scientists A. E.Gileadi, **Electrode Kinetics for Chemists**, Chemical Engineers and Material, VCH 1993.
2. Berry Rice and Ross, **Physical Chemistry published**, 2nd Edition, OUP, 2000.